Note: No Week 12 slides were in the project. Reconstructed from the assignment + the concept-erasure / representation-steering and FORGE (graph) material. The Week 12 screenshot showed **your own selected answers** (not an accepted-answer key), so each answer below is confirmed against the underlying concept — I flag the one worth double-checking.

1 Concept Erasure & Guarded Attributes

This builds on Week 3's Representation Engineering: a **concept/attribute** (e.g. gender, toxicity) lives as a direction in the model's representation space, and we can **erase or steer** it.

'Guarded' attribute (Q1): an attribute is guarded when **it can only be classified by specific linear probes** — i.e. a linear probe can't recover it in general; it's protected against linear classification.

Objective of distribution-matching (Q2): modifying vectors from one distribution to look like another (e.g. toxic → non-toxic) aims **to guard an attribute with the smallest possible change to unrelated semantics** — remove the concept while disturbing everything else as little as possible.

Affine Steering / Affine concept erasure (Q3): its implication is that it is a **no-gradient, cheap way to control model generation** — you apply an affine (linear + shift) transform to activations, no fine-tuning or gradient descent needed.

Assignment: guarded = classifiable only by specific linear probes (Q1); distribution-matching goal = guard an attribute with minimal change to unrelated semantics (Q2); Affine Steering = no-gradient, cheap generation control (Q3).

2 FORGE & Cell Complexes (graph explainability)

Cell complex = a higher-order generalisation of a graph: **0-cells = vertices, 1-cells = edges, 2-cells = cycles/faces**, etc. Lets you model **group-wise (multi-node) interactions** a plain graph can't.

- Cycles are represented as **2-cells** (Q7)
- A conventional cell complex **scales quadratically** with graph size (Q4) — expensive
- A p-cell is a **face/boundary** of a (p+1)-cell if its point-set is a subset of the larger cell's (Q10)

Why a standard GNN explainer fails on molecule toxicity (Q5): it **cannot capture specific multi-node group-wise interactions** (functional groups of atoms acting together).

FORGE (Q6): reduces the cell complex's computational complexity **from quadratic to linear** by **strategically removing specific types of boundary relations** — keeping the useful higher-order structure while dropping the costly parts.

Which methods benefit most from FORGE (Q9): perturbation-based explainer methods (they repeatedly re-evaluate, so a linear-scale complex helps most).

Graph explainers (Q8, select all): Toxicity detection of molecules, Drug prediction, Fraud detection (NOT next-token prediction, NOT affine concept erasure — those aren't graph-explainer tasks).

3 Cell-complex definitions (Q10, select all)

- **True:** a 0-cell corresponds to a vertex, a 1-cell to an edge in traditional graphs
- **True:** a p-cell $e^{(p)}$ is a face/boundary of a (p+1)-cell if its set of points is a **subset** of those composing the larger cell
- **False:** '1-skeleton consists exclusively of 1-D cells' (it also includes the 0-cells); 'vertical boundary relations connect same-dimension cells'; 'boundary relations restricted to undirected'

4 One-line recap for the exam

- **Guarded attribute** = only specific linear probes can classify it
- **Distribution matching** = erase concept, minimal change to unrelated semantics
- **Affine steering** = no-gradient, cheap generation control
- **0/1/2-cells** = vertices / edges / cycles
- **Conventional cell complex** scales **quadratically**; **FORGE** makes it **linear** by removing boundary relations
- **GNN explainer fails** = can't capture multi-node group interactions
- **Perturbation-based** methods gain most from FORGE
- **Graph explainers**: molecule toxicity, drug prediction, fraud detection

Double-check (from your own submission, no answer-key shown): Q4 'scales quadratically' and Q9 'perturbation-based' are the two I'd re-confirm against the lecture if you can — the rest follow directly from the definitions.